

Cloud Resource Graph $G = (V, E, \Phi_V, \Phi_E)$

Node $v \in V$
 Φ_V : provider, tier, HA, $\sigma(v)$

Edge $e \in E$
 Φ_E : kind, proto, $s(e)$

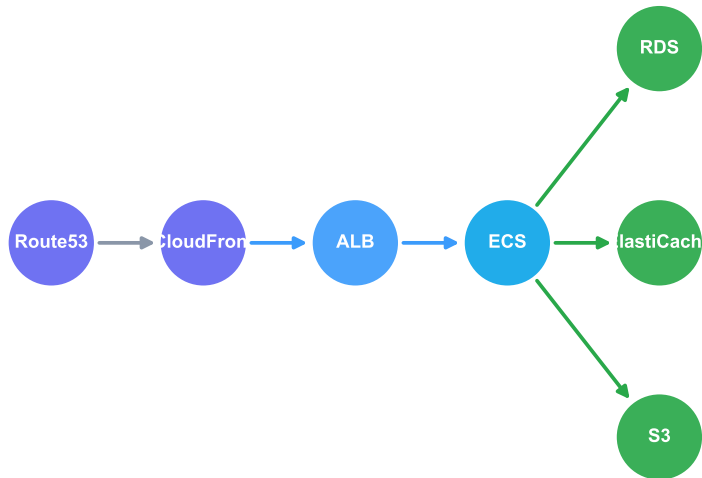
$\sigma(v) = \{\text{auth}, \text{encT}, \text{encR}, \text{pub}, \text{mtls}, \text{waf}, \text{priv}\}$

Trust $\tau(v)$
untrusted / dmz / trusted

Attack surface $A(G)$
public exposure + reachability

Zero-Trust score $S_{\text{ZT}}(G) = \sum w_i c_i(G)$

Three-tier template (planner.py)



Compute

Edge

Ingress

Storage

Data